

BLUE BANDS COLLECTORS

The invoice looks right. Does the payment?

RenderGuard

An evidence-bound release gate for AI-prepared repeat-supplier
EUR payments.

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LIVE DEMO

<https://renderguard.vvitovec.com>

<https://github.com/vvitovec/RenderGuard>

SANDBOX ONLY

No bank connected. No money moved.

THE SPECIFIC FAILURE

One invoice can carry two payment instructions.

User: an accounts-payable operator preparing a repeat-supplier payment against an approved purchase order.

Visible page

Account ...1001

EUR 1,240.00

Hidden PDF text / QR

Account ...9999

Same-looking invoice

Approved supplier

Account ...1001

Independent authority

Payment changes must be independently verified using a saved supplier contact. Document agreement alone cannot authorize a new bank account.

Research: FBI business email compromise guidance; Microsoft vendor-bank approval workflow.

PURPOSE-BUILT WORKFLOW

AI prepares. Independent evidence decides.

01	02	03	04	05
Select authority	Establish evidence	Constrain the AI	Approve exact intent	Release once
Approved repeat supplier + PO	Raster / OCR / PDF text / EPC QR	Semantic guard + bank handles	Reviewer-bound action + evidence	Protected sandbox ledger receipt

Bank verification is a separate administrator workflow. The assistant cannot change supplier authority, approve itself or execute a payment.

A COMPLETE POSITIVE PATH

A legitimate invoice reaches a saved receipt.

Actual rendered PDF -> local model proposal -> human reviewer -> immutable sandbox ledger.

The AI receives opaque account handles and evidence-grounded structured fields.

The reviewer sees the exact recipient, EUR amount, invoice and approved obligation.

Execution rechecks authority and records one receipt. A retry returns that same receipt.

RenderGuard

Workbench

Controls

Release register

Test lab

How it works

SANDBOX

Reviewer

Local AI connected

Demo personas share your fictional workspace; production requires external identity.

New workspace

ACCOUNTS PAYABLE / RELEASE REVIEW

Payment workbench

Release only what the invoice, approved obligation and supplier authority agree on.

Add invoice

REVIEW QUEUE1

Clean

P0-2609-014

Awaiting reviewer

DOCUMENT EVIDENCE

NF-2026-104

Evidence ready

Rendered pagePayment fieldsPDF & AI input

Independent raster → OCR · 753 ms

Nordlicht Facilities GmbH

FICTIONAL SUPPLIER · HACKATHON TEST DOCUMENT

INVOICE NF-2026-104

Bill to: Alder Works s.r.o.
Issue date: 01 October 2026
Due date: 14 October 2026
Purchase order: PO-2609-014

SERVICE

AMOUNT

Quarterly facilities maintenance

1,240.00

VAT: reverse charge / test document

RELEASE DECISION

Released to sandbox

Release recorded.

An immutable receipt exists in the sandbox ledger.

SupplierNordlicht Facilities GmbH

Payment€1,240.00

Recipient accountDE5999999999000001001

Approved obligationP0-2609-014

AI PROPOSAL RATIONALE

PAYMENT INSTRUCTIONS Beneficiary: Nordlicht Facilities GmbH IBAN: account_1
Reference: NF-2026-104 SEPA payment QR Scan to prefill recipient and amount.

✓

RG-60846765D313

Sandbox only · no bank connected
944042e02fd825b0d74c4209...

Open release register →

Blue Bands Collectors / HackYeah 2026 / English

FINAL - PREPARED FOR REVIEW, NOT SUBMITTED

CONCRETE ADVERSARIAL EVIDENCE

The page and the payment QR disagree.

The actual QR-swap PDF is blocked before any model dispatch.

Visible recipient: ...1001. Decoded EPC
recipient: ...9999. Approved supplier:
...1001.

Also tested: hidden text recipient / total, QR
amount swap, unverified account change,
ambiguity and overbilling.

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How it works

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Operator

Local AI connected

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New workspace

ACCOUNTS PAYABLE / RELEASE REVIEW

Payment workbench

Release only what the invoice, approved obligation and supplier authority agree on.

Add invoice

REVIEW QUEUE1

Qr Swap

P0-2609-014

Evidence ready

DOCUMENT EVIDENCE

NF-2026-104

Evidence ready

Rendered page	Payment fields	PDF & AI input
Does every representation agree?		
SOURCE	RECIPIENT	TOTAL
VISIBLE / OCR	DES9999999999000001001	€1,240.00
PDF TEXT	DES9999999999000001001	€1,240.00
PAYMENT QR 1	DE98999999990000009999	€1,240.00
APPROVED SUPPLIER	DES9999999999000001001	€1,240.00

Supplier authority is selected before the invoice is read. An invoice cannot approve its own new account.

Visible payment instructions

Passed

OCR evidence quality

Passed

RELEASE DECISION

Blocked

Keep this payment on hold.

QR differs from visible recipient.

Supplier

Nordlicht Facilities GmbH

Payment

€1,240.00

Recipient account

DES9999999999000001001

Approved obligation

P0-2609-014

Checking local AI...

Approval is invalidated by a change in payment, source, rendered evidence, supplier authority or policy.

Visible payment instructions

Passed

OCR evidence quality

Passed

HYBRID CONTROL LAYER

Deterministic authority. Semantic interpretation.

DETERMINISTIC RELEASE CONTROLS

Identity, models, tools, privacy patterns, literal attack signatures, evidence agreement, supplier/PO, action schema and budget reservations.

REAL LOCAL SEMANTIC GUARD

Qwen2.5:3b classifies behavioral overrides in untrusted document content. The proposal model is constrained to evidence-grounded fields.

Policy and literal signature catalog can be edited live. Historical probes inspect CVE-2025-6514 unsafe MCP authorization endpoints and unsafe pickle formats; they never execute supplied code.

PROTECTED EXECUTION

An approval means one exact action.

Payment hash

Source PDF + rendered-image hashes

Structured evidence hash

Approved supplier + obligation hashes

Policy version + signature catalog

Workspace, reviewer capability and expiry

A change invalidates the approval.

A single transaction revalidates, consumes approval and inserts the receipt. Unique proposal / invoice constraints prevent duplicate effects.

8 concurrent retries -> 1 receipt

Executable concurrency test; actual SQLite ledger.

Show the decision. Account for the cost.

Actual decision trail, policy versions, model tokens, stage latency, bound hashes and immutable receipts.

Export redacted JSONL and sandbox release CSV. Approval tokens never enter exports.

Reserve call / token / cost / concurrency limits before dispatch. Unknown interrupted usage consumes its reservation.

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Workbench

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How it works

SANDBOX

Reviewer

Local AI connected

Demo personas share your fictional workspace; production requires external identity.

New workspace

A sandbox release receipt was saved. No bank is connected and no money moved.

REPORTING / ACTUAL WORKSPACE EVENTS

Release register

Inspect decisions, budget consumption and immutable sandbox receipts.

Refresh

1 sandbox releases

16 audit events

0 blocked events

875 accounted tokens

7228 ms stage p95

Event counts include gateway and workflow stages; they are not unique invoices or claimed fraud savings. Latency covers completed model / document stages in this workspace.

Immutable sandbox receipts

Export release CSV

RECEIPT	INVOICE / SUPPLIER	AMOUNT	EXACT ACTION HASH	RELEASED
RG-60846765D313	NF-2026-104 nordlicht	€1,240.00	944042e02fd825b0d74c...	16:03:50

Decision trail

Export redacted JSONL

16:03:50	payment.released	Verified
16:03:49	payment.approved	Verified

REPRODUCIBLE EVIDENCE

Positive cases matter as much as attacks.

Finite synthetic corpus and stated prompts. No universal fraud or prompt-injection guarantee.

83/83

16/16

6/6

8/8

Control / integration checks

Full live PDF workflows

Real semantic prompt checks

Headless browser workflows

Actual OCR + labelled provider double

Real local AI, worker, approval, release

Benign terms and behavioral overrides

No mocked network; desktop + mobile

Evidence expectations: 16/16. Full workflow: seven legitimate variants + nine blocked cases.

Recorded reports include source-content hashes; commands and expected cases are packaged.

READY TO INSPECT AND CHANGE

A complete gate, with an explicit boundary.

Robustness: exact protected release and adversarial checks.

Architecture: isolated worker, modular gateway and provider. Reporting: actual audit and export. Tests: executable positive and negative suite.

Scale: bounded resources and a documented migration path.

Deployed on always-on hosts. No paid key required. Team and submission materials are fully English.

<https://renderguard.vvitovec.com>

SUPPORTED

Repeat supplier / approved EUR obligation / labelled PDF / EPC QR / sandbox ledger.

PRODUCTION INTEGRATION

External identity, independently managed ERP authority, disposable workers and an idempotent bank adapter.